



Anomaly Detection for Fraud Risk in Banking Data





Banking Transaction

City

All

2020

2021

2022

2023

2024

2025

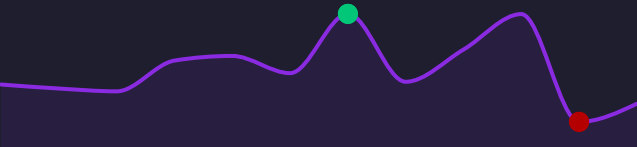


Total Amount

PY: \$3.09M

\$2.40M

-22.24% ▼ vs 2023



Total Tracsaction

PY: 10.44K

7.97K

-23.62% ▼ vs 2023



Total Customer

PY: 495

493

-0.40% ▼ vs 2023

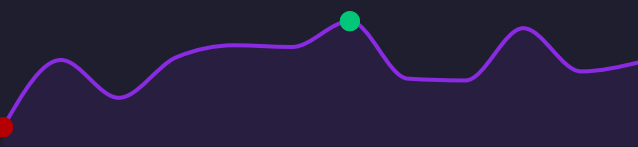


Avg Trans Value

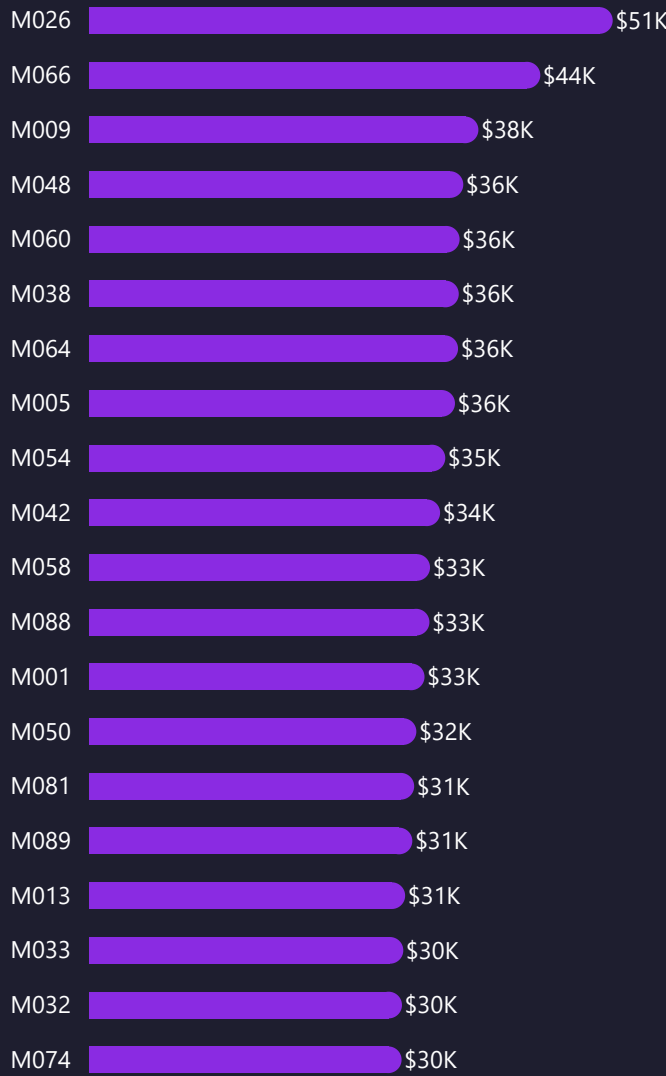
PY: \$295.87

\$301.20

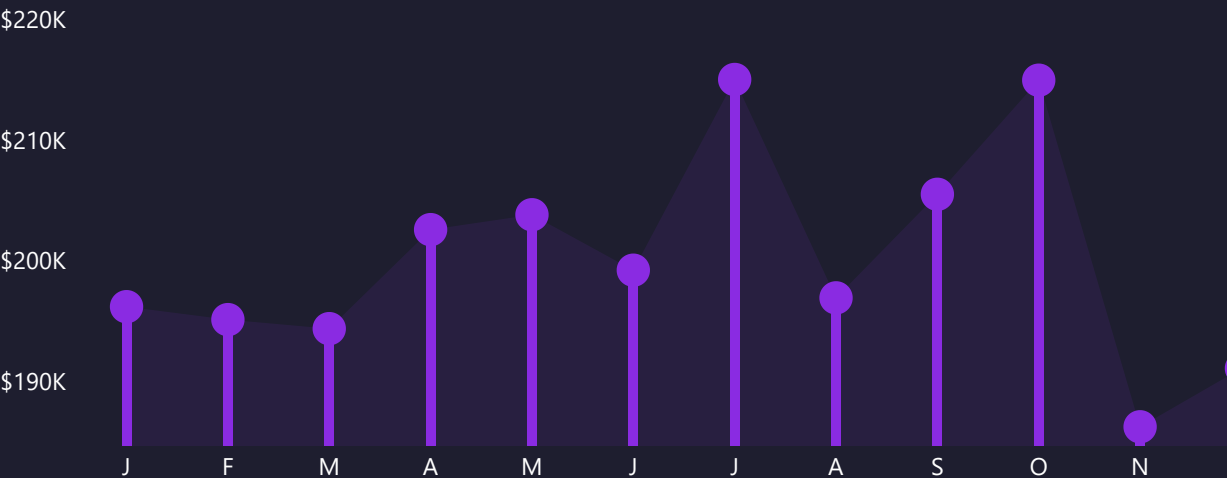
1.80% ▲ vs 2023



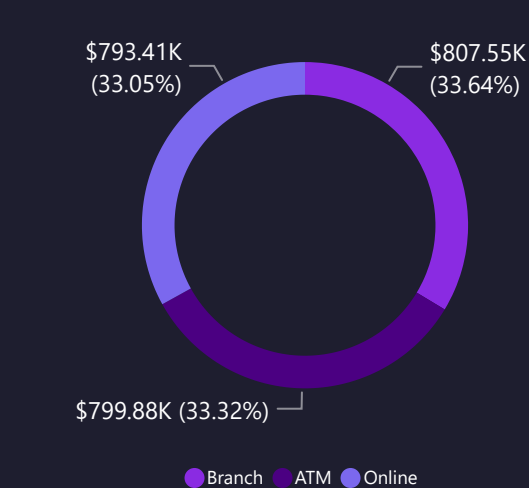
Top 20 Amount by Merchant



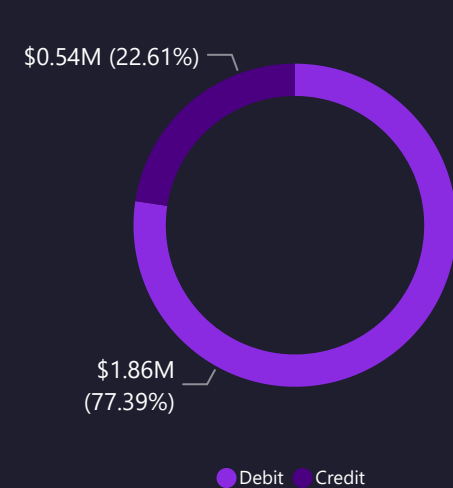
Amount by Month



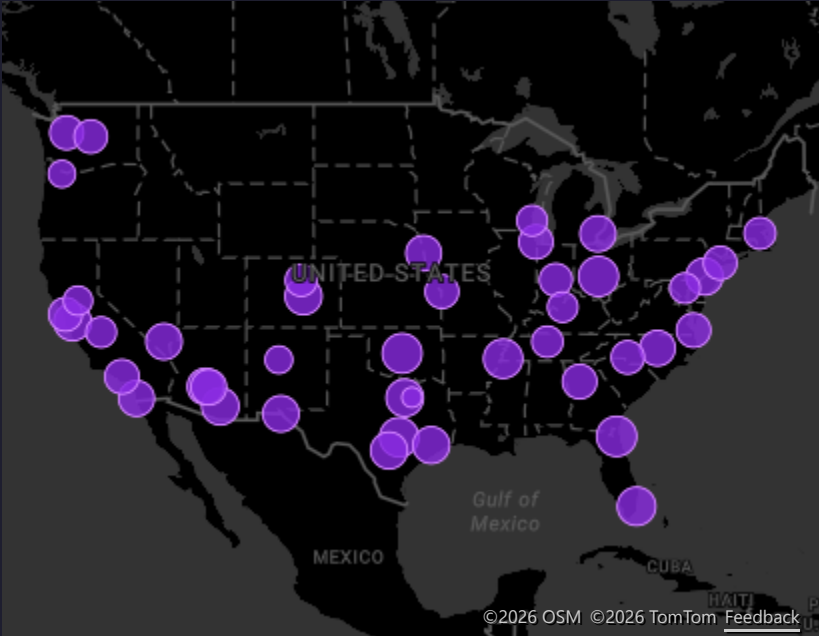
Amount by Channel



Amount by TransactionType



Amount by Location



City [Top 5]	Amount %	Total Amount
Jacksonville	20.57%	\$77,912.00
Columbus	20.32%	\$76,967.00
Oklahoma City	20.06%	\$75,987.00
Austin	19.59%	\$74,203.00
Memphis	19.47%	\$73,774.00



Banking Transaction

MTD

QTD

YTD

City

All

2020

2021

2022

2023

2024

2025

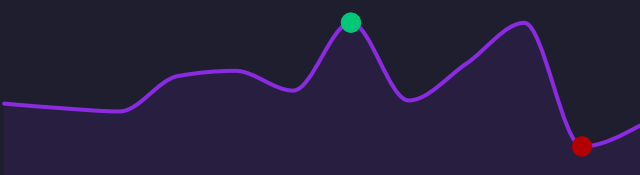


Amount MTD

PY: \$259.74K

\$191.07K

-26.44% ▼ vs 2023



Transaction MTD by Channel

PY: 877

633

-27.82% ▼ vs 2023

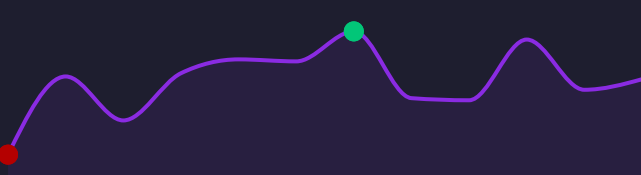


Avg Trans Value MTD

PY: \$296.17

\$301.84

1.92% ▲ vs 2023

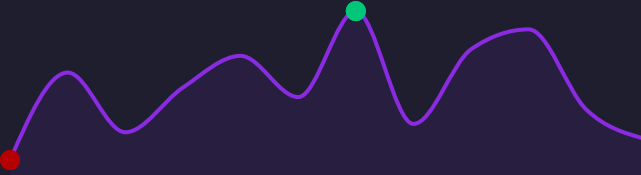


Avg Cus Value MTD

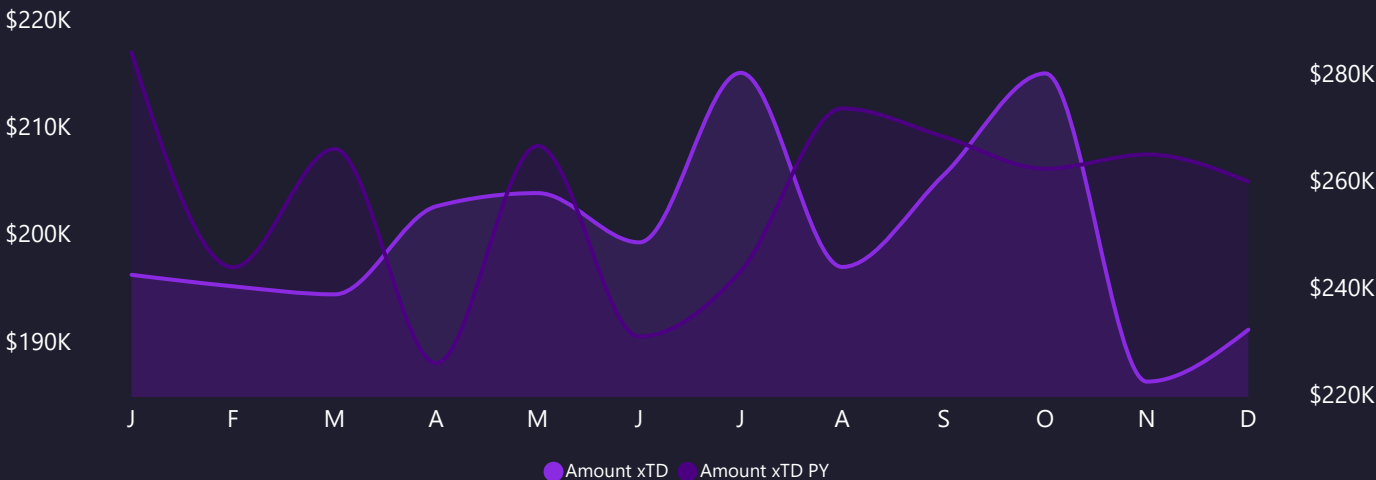
PY: \$664.29

\$557.04

-16.15% ▼ vs 2023



Amount MTD and Amount MTD PY by Month



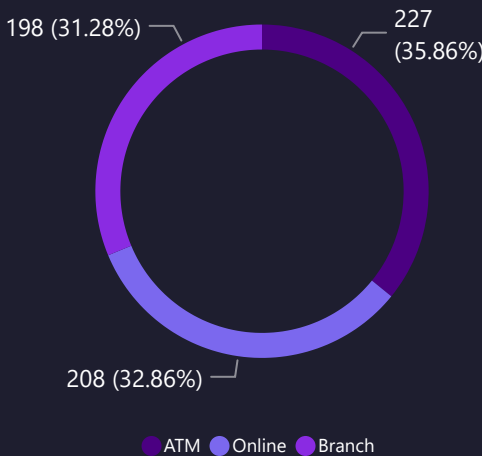
Top 20 ACV MTD by Customer

AC00312	\$3.5K
AC00111	\$3.0K
AC00202	\$2.8K
AC00416	\$2.7K
AC00217	\$2.3K
AC00164	\$2.1K
AC00150	\$2.0K
AC00363	\$2.0K
AC00068	\$1.9K
AC00128	\$1.9K
AC00290	\$1.9K
AC00179	\$1.9K
AC00297	\$1.8K
AC00257	\$1.7K
AC00396	\$1.7K
AC00237	\$1.6K
AC00192	\$1.5K
AC00317	\$1.5K
AC00366	\$1.5K
AC00182	\$1.5K
AC00482	\$1.5K

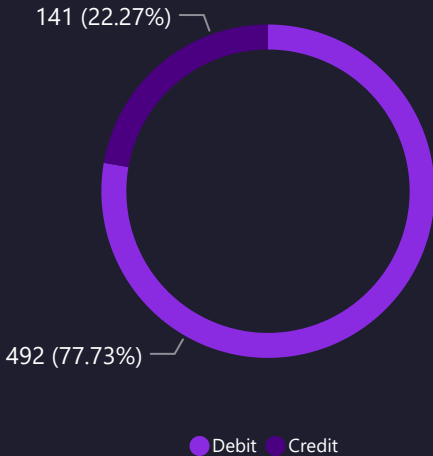
Top 10 Amount MTD by Device

D000106	\$3.1K
D000344	\$3.0K
D000103	\$3.0K
D000017	\$2.9K
D000674	\$2.8K
D000205	\$2.5K
D000312	\$2.1K
D000443	\$2.0K
D000037	\$1.9K
D000400	\$1.9K

Transaction MTD by Channel



Transaction MTD by Transaction Type



City [Top 7]	Trans Growth % xTD
Sacramento	-2.13%
Raleigh	-3.70%
San Diego	-6.67%
Oklahoma City	-10.94%
Columbus	-12.90%
El Paso	-15.69%
Indianapolis	0.00%



Banking Transaction

Amount

Account Balance

City

All

2020

2021

2022

2023

2024

2025

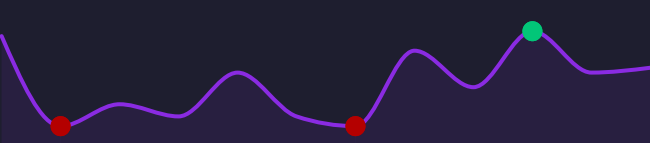


Total Customer

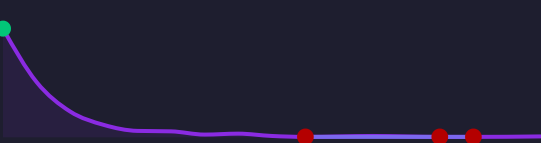
495

PY: 494

0.20% ▲ vs 2022



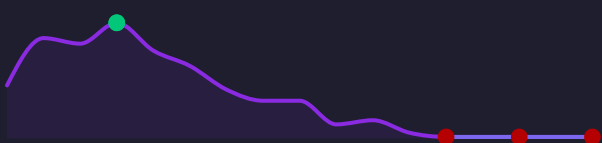
R Average Recency
29



F Average Frequency
101



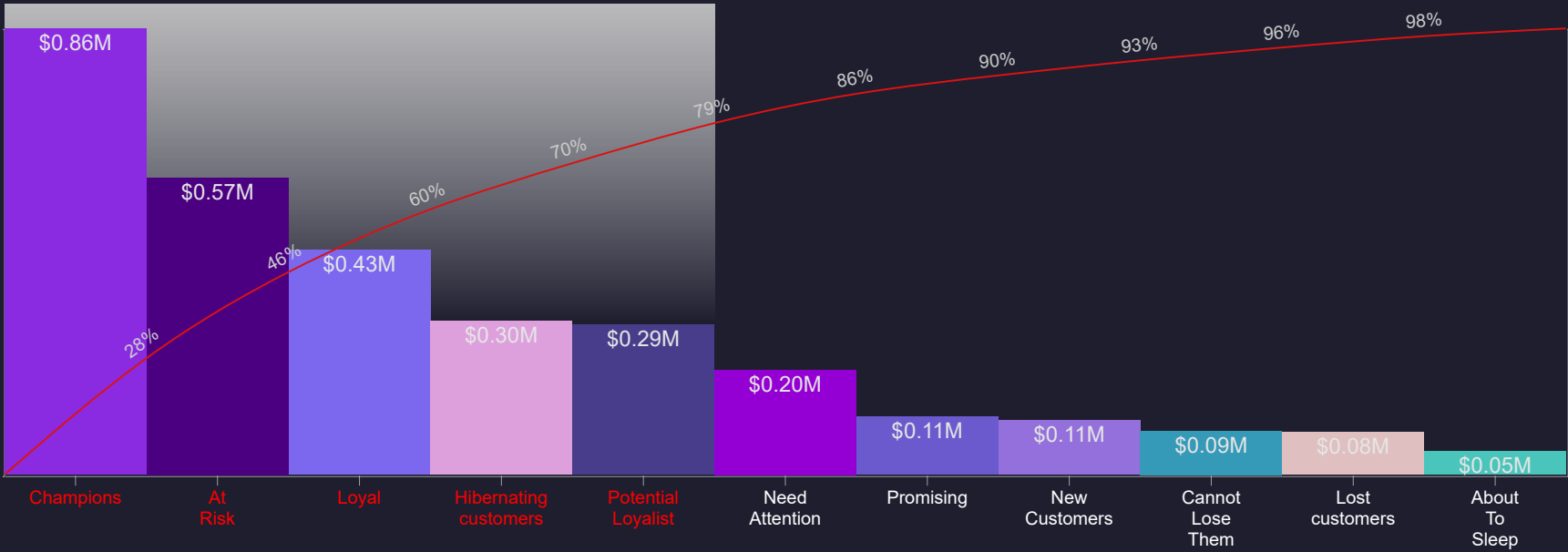
M Average Monetary
\$30.09K



Total Customer by Segment Customer



Pareto Analysis of Customer Segments



AccountID	Segment Customer	RFM_Score	Recency	Frequency	Monetary
AC00014	Champions	555	3	176	\$64,202
AC00030	Champions	555	4	184	\$61,908
AC00032	Champions	555	3	187	\$72,499
AC00035	Champions	555	4	190	\$48,228
AC00071	Champions	555	5	157	\$66,602
AC00086	Champions	555	2	146	\$62,681
AC00095	Champions	555	3	179	\$81,857
AC00136	Champions	555	1	159	\$77,692



All



First Transaction Month	2020-Jan	2020-Feb	2020-Mar	2020-Apr	2020-May	2020-Jun	2020-Jul	2020-Aug	2020-Sep	2020-Oct	2020-Nov	2020-Dec	2021-Jan	2021-Feb	2021-Mar	2021-Apr	2021-May	2021-Jun	2021-Jul	2021-Aug
01/01/2020	350	262	261	265	264	255	248	269	259	270	261	279	264	235	266	261	261	253	254	269
02/01/2020		79	50	48	53	47	49	49	43	42	52	57	51	44	45	52	46	52	53	61
03/01/2020			33	20	20	20	14	21	17	21	17	20	18	16	20	15	17	17	18	19
04/02/2020				14	7	6	10	7	7	5	9	3	9	5	8	7	4	7	7	4
05/04/2020					7	2	5	3	3	4	4	1	2	3	3	4	3	3	2	4
06/04/2020						4	3	2	3	1		2	4	2	1	3	3	2	3	2
07/09/2020							3		1	1			1		2	1	1	2	1	3
11/09/2020											1									
12/23/2020												2		1	1	1	1	1	1	
02/09/2021														1	1					
04/08/2021																1				1
Total	350	341	344	347	351	334	332	351	333	344	344	364	349	307	347	345	336	337	339	363

First Transaction Month	2020-Jan	2020-Feb	2020-Mar	2020-Apr	2020-May	2020-Jun	2020-Jul	2020-Aug	2020-Sep	2020-Oct	2020-Nov	2020-Dec	2021-Jan	2021-Feb	2021-Mar	2021-Apr	2021-May	2021-Jun	2021-Jul	2021-Aug
01/01/2020	100.00%	74.86%	74.57%	75.71%	75.43%	72.86%	70.86%	76.86%	74.00%	77.14%	74.57%	79.71%	75.43%	67.14%	76.00%	74.57%	74.57%	72.29%	72.57%	76.86%
02/01/2020		100.00%	63.29%	60.76%	67.09%	59.49%	62.03%	62.03%	54.43%	53.16%	65.82%	72.15%	64.56%	55.70%	56.96%	65.82%	58.23%	65.82%	67.09%	77.22%
03/01/2020			100.00%	60.61%	60.61%	60.61%	42.42%	63.64%	51.52%	63.64%	51.52%	60.61%	54.55%	48.48%	60.61%	45.45%	51.52%	51.52%	54.55%	57.58%
04/02/2020				100.00%	50.00%	42.86%	71.43%	50.00%	50.00%	35.71%	64.29%	21.43%	64.29%	35.71%	57.14%	50.00%	28.57%	50.00%	50.00%	28.57%
05/04/2020					100.00%	28.57%	71.43%	42.86%	42.86%	57.14%	57.14%	14.29%	28.57%	42.86%	42.86%	57.14%	42.86%	42.86%	28.57%	57.14%
06/04/2020						100.00%	75.00%	50.00%	75.00%	25.00%		50.00%	100.00%	50.00%	25.00%	75.00%	75.00%	50.00%	75.00%	50.00%
07/09/2020							100.00%		33.33%	33.33%			33.33%		66.67%	33.33%	33.33%	66.67%	33.33%	100.00%
11/09/2020											100.00%									
12/23/2020												100.00%		50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	
02/09/2021														100.00%	100.00%					
01/09/2021																100.00%				100.00%



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Avg Transaction Value

\$295.87

Purchase Frequency

21.08

GrossMargin

30.00%

Churn Rate

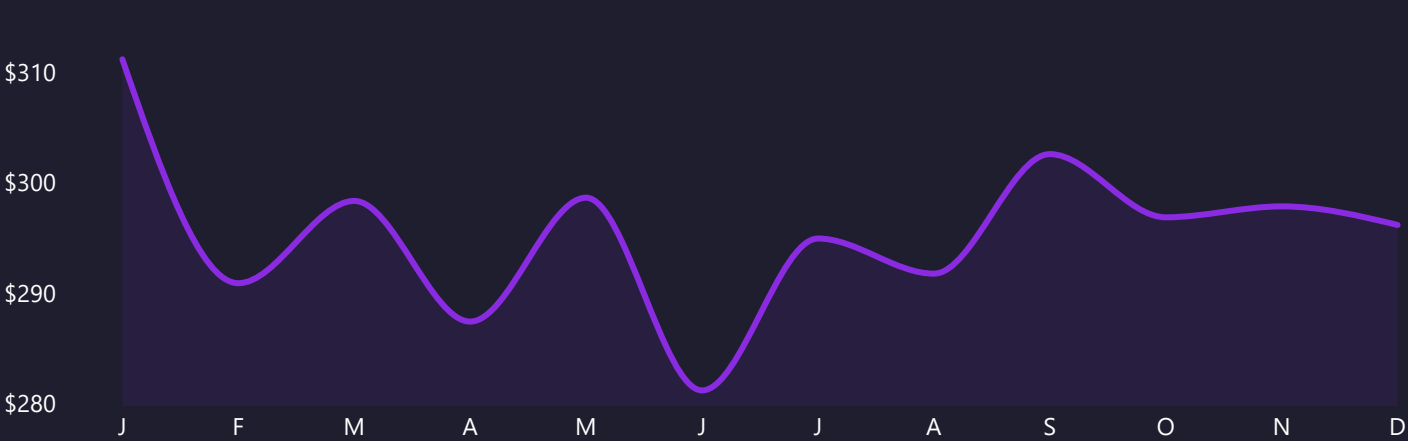
0.20%

Customer Lifetime Value CLV

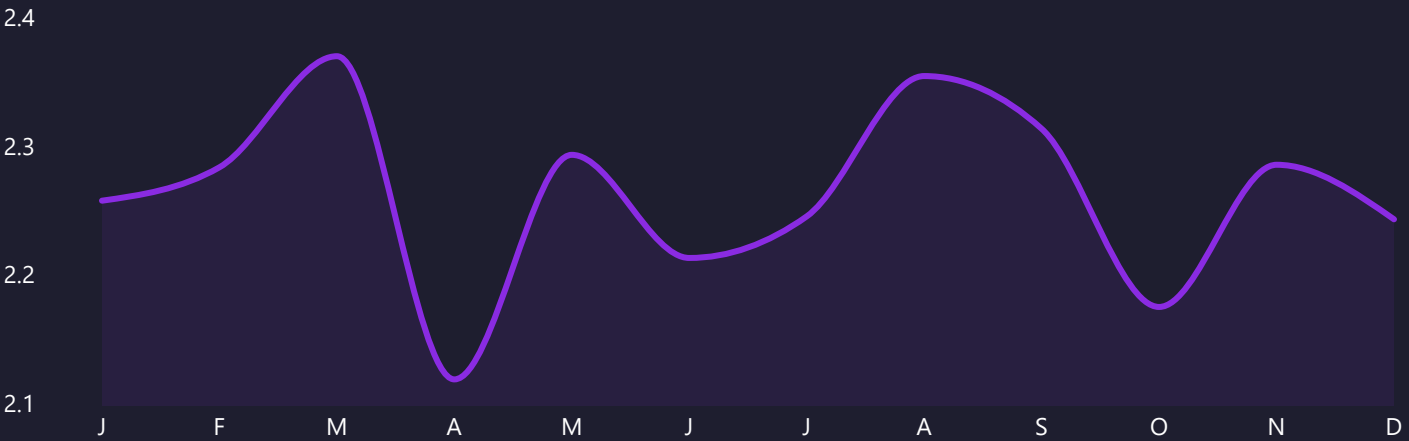
\$926.30K



ATV by Month



Purchase Frequency by Month



Customer Lifecycle Metrics by Month

Month	Total Customer	Total Tracsaction	Purchase Frequency	Avg Transaction Value	Total Account Balance	Total Trans Amount	ATV	GrossMargin
Jan	404	912.00	2.26	\$311.18	\$4,471,479.25	\$283,798.00	\$311.18	30.00%
Feb	367	838.00	2.28	\$290.89	\$4,276,003.82	\$243,765.00	\$290.89	30.00%
Mar	376	891.00	2.37	\$298.35	\$4,498,885.71	\$265,826.00	\$298.35	30.00%
Apr	371	786.00	2.12	\$287.41	\$4,077,352.8	\$225,903.00	\$287.41	30.00%
May	389	892.00	2.29	\$298.64	\$4,475,539.51	\$266,387.00	\$298.64	30.00%
Jun	371	821.00	2.21	\$281.17	\$4,207,734.21	\$230,842.00	\$281.17	30.00%
Jul	367	824.00	2.25	\$294.94	\$4,151,977.95	\$243,030.00	\$294.94	30.00%
Aug	398	937.00	2.35	\$291.75	\$4,676,292.53	\$273,374.00	\$291.75	30.00%
Sep	383	886.00	2.31	\$302.59	\$4,610,078.15	\$268,094.00	\$302.59	30.00%
Oct	406	883.00	2.17	\$296.86	\$4,645,969.87	\$262,128.00	\$296.86	30.00%
Nov	389	889.00	2.29	\$297.85	\$4,480,587.82	\$264,791.00	\$297.85	30.00%
Dec	391	877.00	2.24	\$296.17	\$4,645,922.62	\$259,739.00	\$296.17	30.00%





Banking Transaction

2020

2021

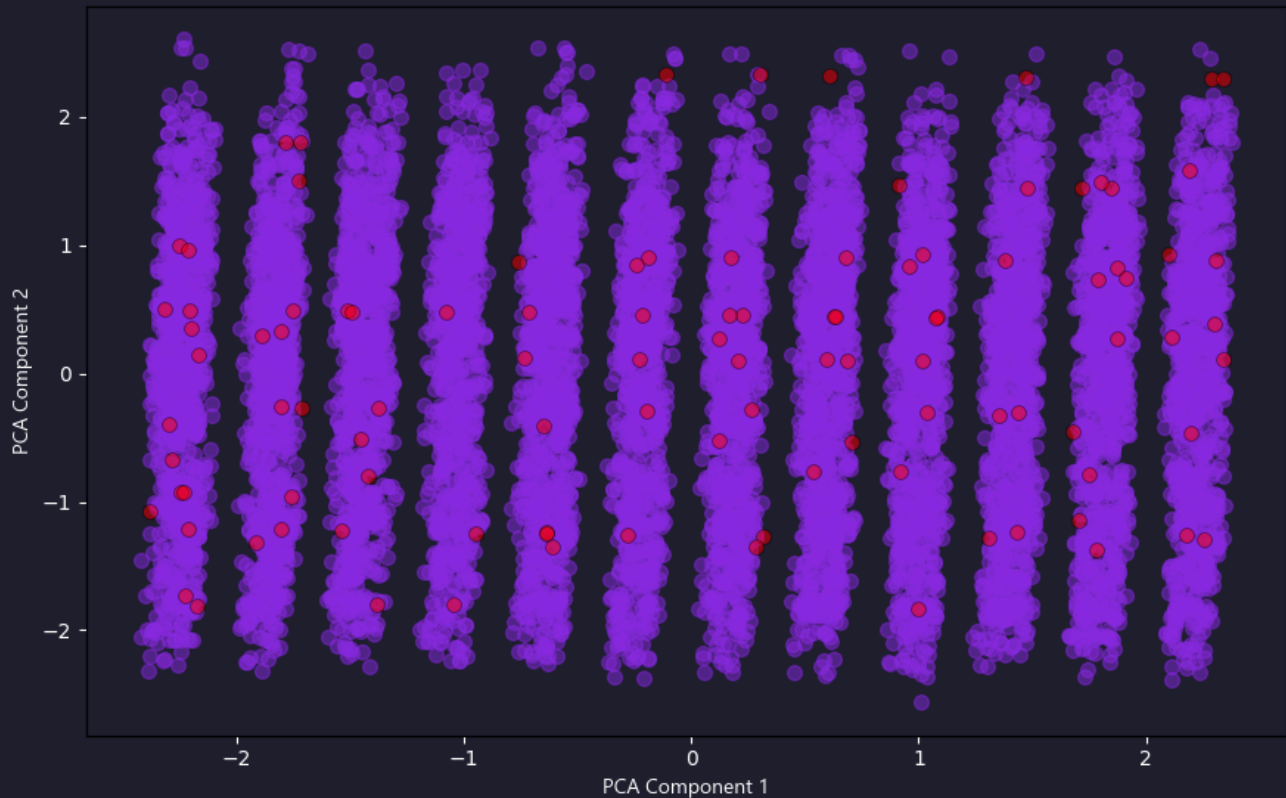
2022

2023

2024

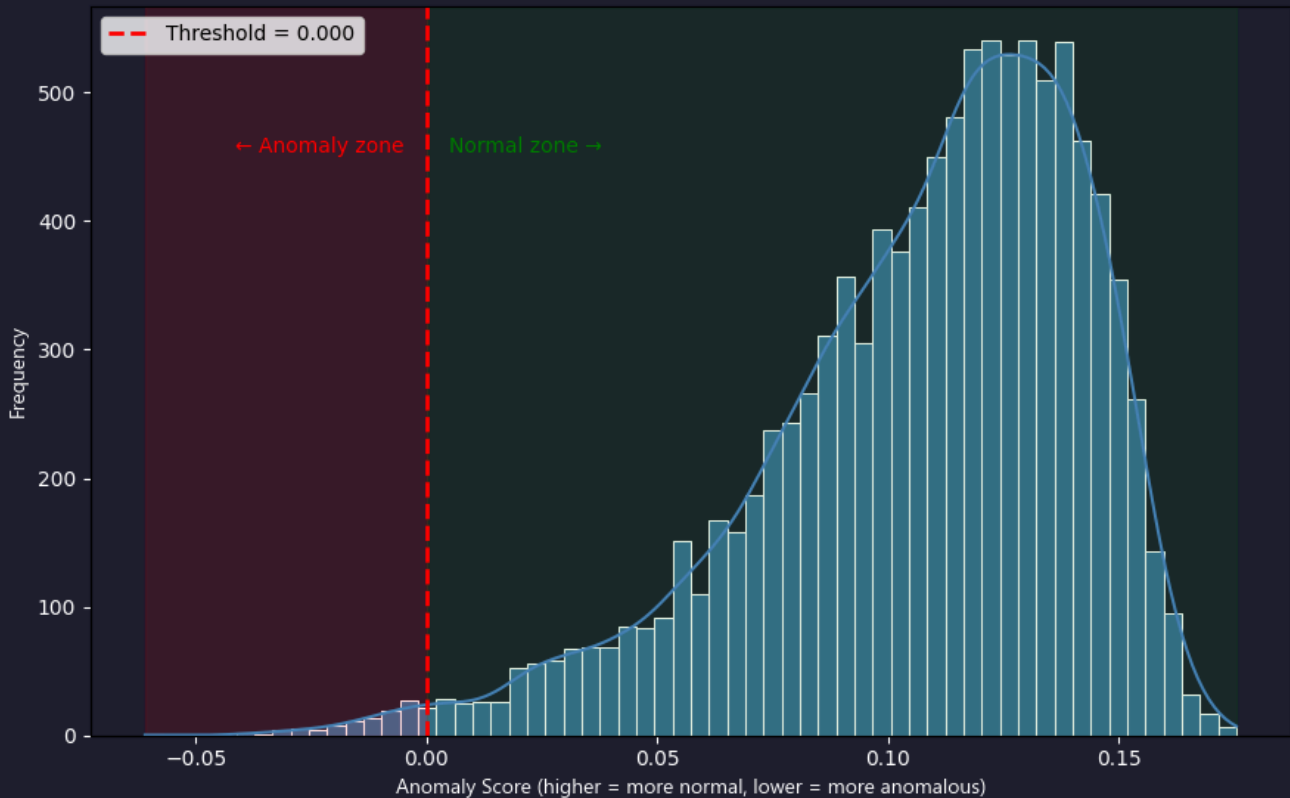
2025

Isolation Forest — Potential Frauds vs Normal (PCA 2D)



Prediction
● Normal (n=10331)
● Potential Fraud (n=105)

Distribution of Isolation Forest Anomaly Scores



TransactionID	AccountID	CustomerAge	CustomerOccupation	TransactionAmount	Location	Channel	TransactionType	TransactionDuration	AnomalyLabel	Segment Customer
TX005872	AC00071	18	Student	\$132	Mesa	Branch	Debit	22	Potential Fraud	Champions
TX000509	AC00353	19	Student	\$127	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX001056	AC00317	19	Student	\$13	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX006101	AC00353	19	Student	\$130	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX009140	AC00353	19	Student	\$115	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX014111	AC00353	19	Student	\$129	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX016718	AC00353	19	Student	\$126	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX022115	AC00317	19	Student	\$13	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX029241	AC00353	19	Student	\$122	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX040164	AC00317	19	Student	\$13	Las Vegas	Online	Debit	20	Potential Fraud	Champions



Banking Transaction

2020

2021

2022

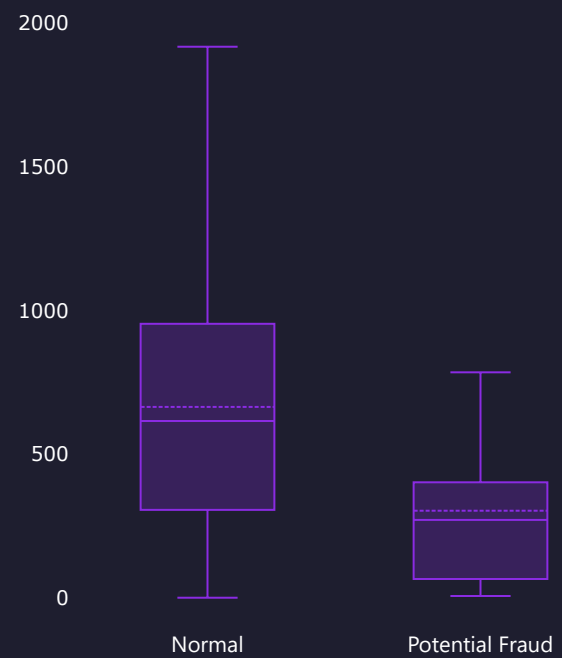
2023

2024

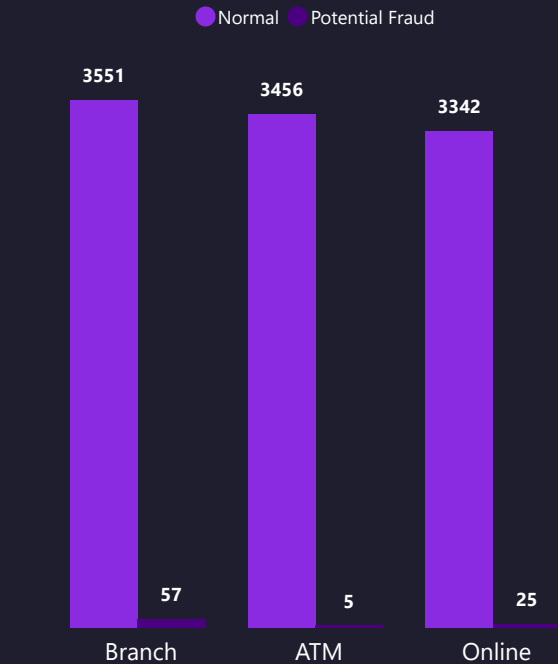
2025



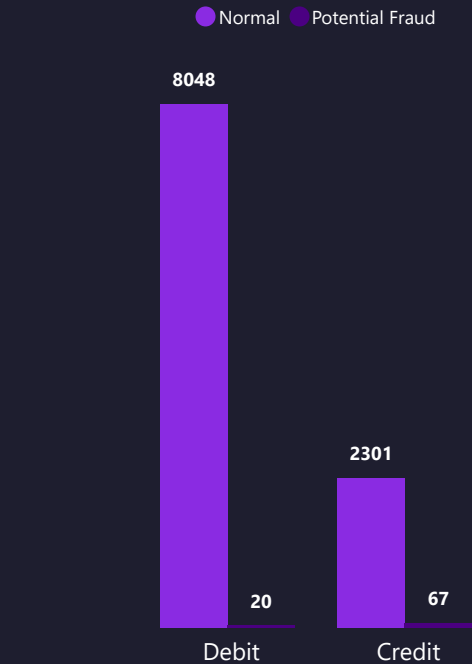
Trans Amount Distribution by Anomaly Label



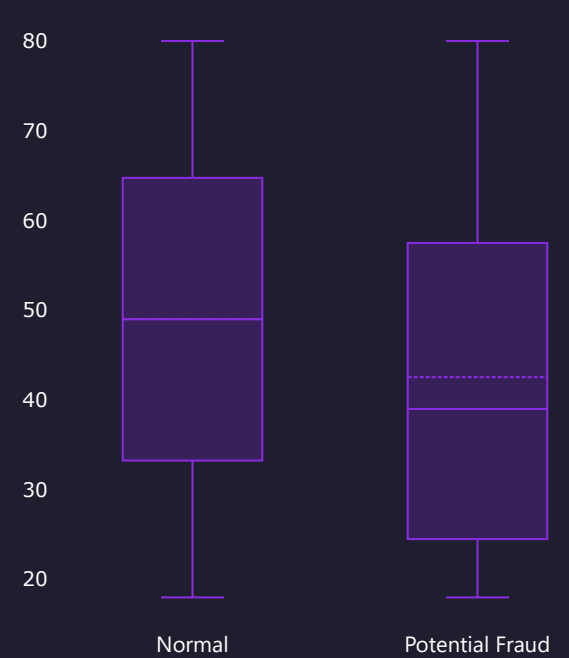
Fraud Pattern by Channel



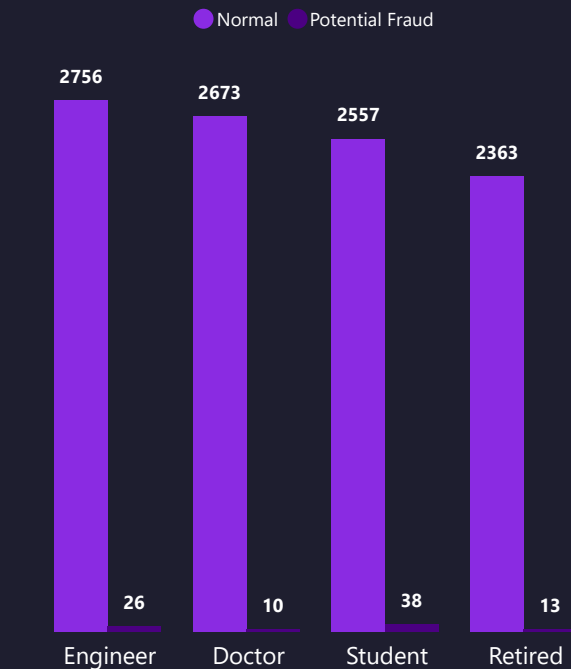
Fraud Pattern by Transaction Type



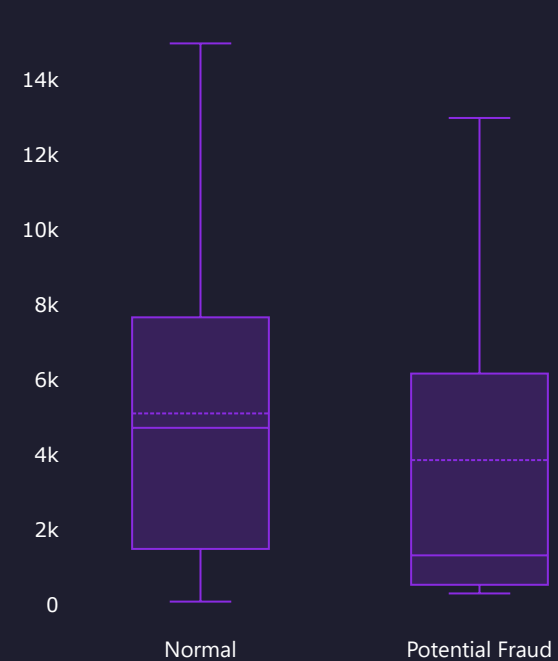
Customer Age Distribution by Anomaly Label



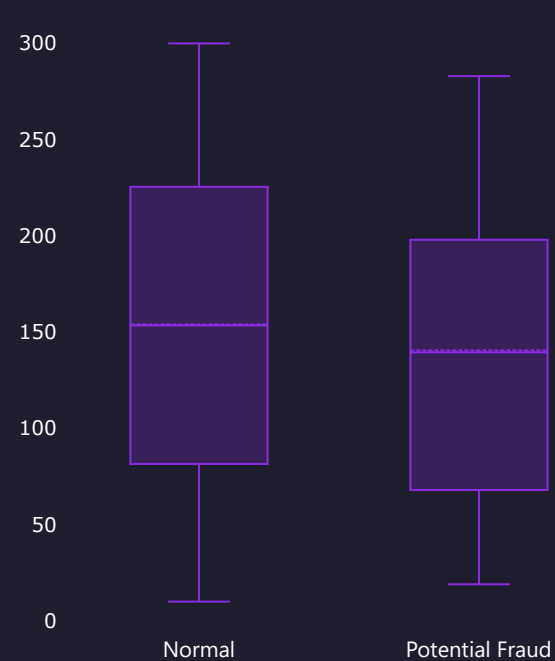
Fraud Pattern by Customer Occupation



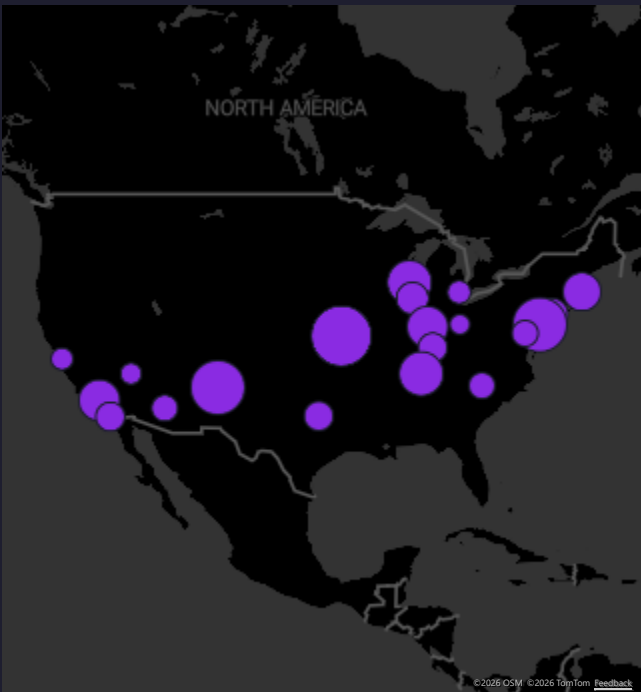
Acc Balance Distribution by Anomaly Label



Trans Duration Distribution by Anomaly Label



Geographic Distribution of Fraud





Banking Transaction

2020

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Monthly Trend of Average Transaction Amount by Anomaly Label

